

AI CALLING AGENT

LLM Evaluation & Performance Report

Based on the Master Evaluation Evidence Workbook

Summary:

Evaluation scale - The master evidence contains 15 unique model-evaluation runs, 131 customer turns and 520 actual model-turn evaluations. The current four-model comparison accounts for 480 of those observations; 40 observations belong to historical/out-of-scope model identifiers retained for traceability.

Controlled benchmark - The final controlled benchmark evaluated the four current models over the same conversation sequence: 48 expected and 48 evaluated observations, with no skipped evaluations. All four achieved 100% grounded-response rate and 0% unsupported-claim rate in that benchmark.

Latency - Aggregate LLM response latency across the accumulated evidence ranges from 4.06 to 7.69 seconds on average for the four current models. The final English benchmark itself averaged 8.46–12.80 seconds. These are model response times only; STT, TTS, Twilio, network and infrastructure are excluded.

Interpretation - The evaluation framework is now measuring quality, grounding, conversation behavior, language behavior, latency, tokens and estimated LLM cost across a meaningful accumulated dataset. Production voice performance remains a separate validation step.

1. Evaluation Scale & Evidence Base

Evidence measure	Count
Unique model-evaluation runs	15
Customer turns	131
Model-turn evaluations	520
Labelled evaluations	412
Unlabelled turns	108
Models represented in evidence	6
Languages represented	4
Framework tests (separate)	149

Framework tests are intentionally excluded from model-trial counts. A model-turn evaluation means one actual model response evaluated for one customer turn.

2. Evaluation Methodology

The evaluation compares model responses against expected intents/actions and evaluates factuality, grounding, conversation quality, language behavior, latency and token usage. The evidence is primarily offline; Twilio is disabled in the recorded evaluation runs.

- Controlled benchmark: fixed scenarios with defined expectations.
- Exploratory interactive evaluation: manually entered customer sequences evaluated by the same model-comparison framework.
- Framework validation: automated tests of evaluators, adapters, interactive session handling and multilingual infrastructure.
- Run-level deduplication: exported copies of the same RUN_ID are not counted as additional trials.

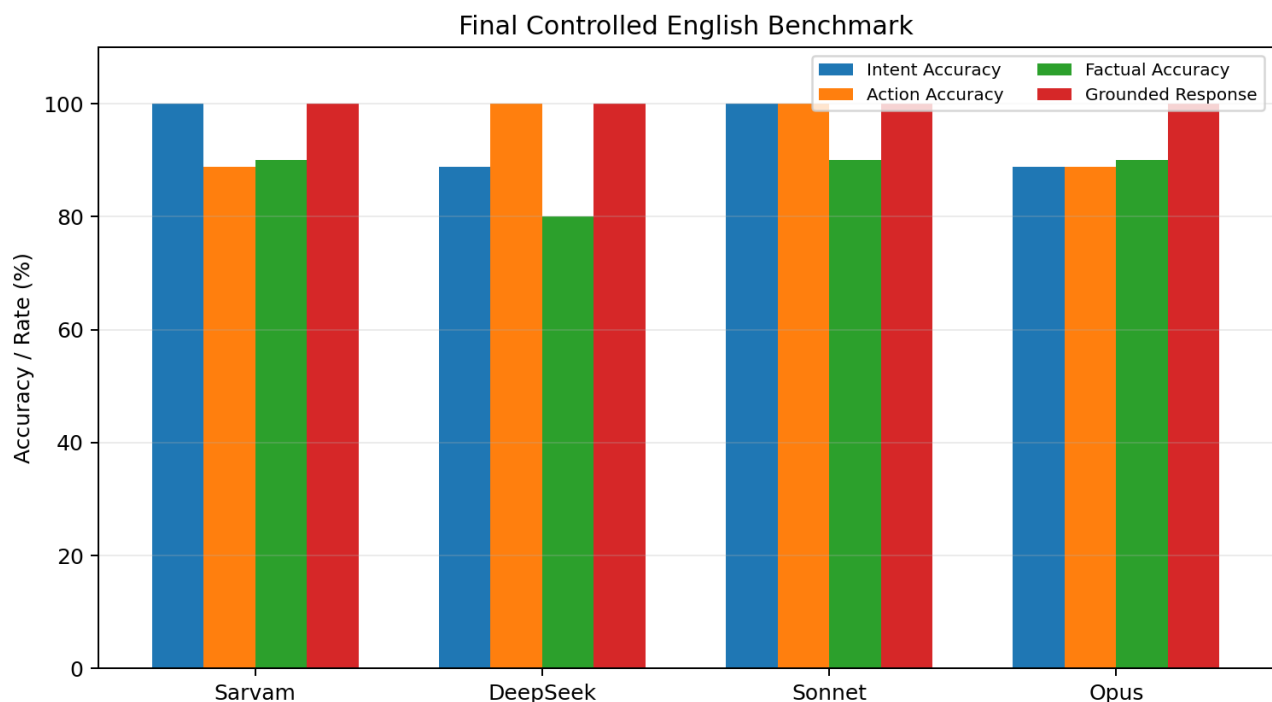
3. Models Evaluated

Model	Provider	Reporting treatment
Sarvam 105B Conversations	Sarvam	Current comparison
DeepSeek V4.1 Flash	DeepSeek	Current comparison
Claude Sonnet 4.6	Anthropic	Current comparison
Claude Opus 4.8	Anthropic	Current comparison
Sarvam Flagship	Sarvam	Historical/out-of-scope evidence
claude-sonnet-4-20250514	Anthropic	Historical/out-of-scope evidence

4. Final Controlled English Benchmark

RUN_ID 20260923_084857 is the primary controlled English comparison. It contains 48 model-turn observations, 12 for each current model, using the same customer sequence.

Model	N	Intent	Action	Factual	Grounded	Unsupported Claims	Avg Latency
Sarvam	12	100%	89%	90%	100%	0%	8.46s
DeepSeek	12	89%	100%	80%	100%	0%	8.98s
Claude Sonnet	12	100%	100%	90%	100%	0%	12.80s
Claude Opus	12	89%	89%	90%	100%	0%	10.73s



5. Aggregate Model Performance

Aggregate metrics below are calculated from the accumulated master evidence. Sample sizes differ by model, so the N column should always be read alongside the percentages. No overall score or ranking is used.

Model	N	Intent	Action	Factual	Grounded	Context	Stage
Sarvam	135	83.2%	80.4%	88.5%	98.5%	79.6%	76.8%
DeepSeek	95	86.6%	92.5%	81.8%	99.0%	81.0%	75.8%
Claude Sonnet	125	83.5%	89.7%	89.7%	96.0%	82.2%	67.4%
Claude Opus	125	87.0%	90.9%	87.9%	99.1%	81.4%	68.4%

The workbook also contains precision, recall, macro-F1, relevance, completeness, clarity, conversational quality and naturalness at model level.

6. Error & Behavior Analysis

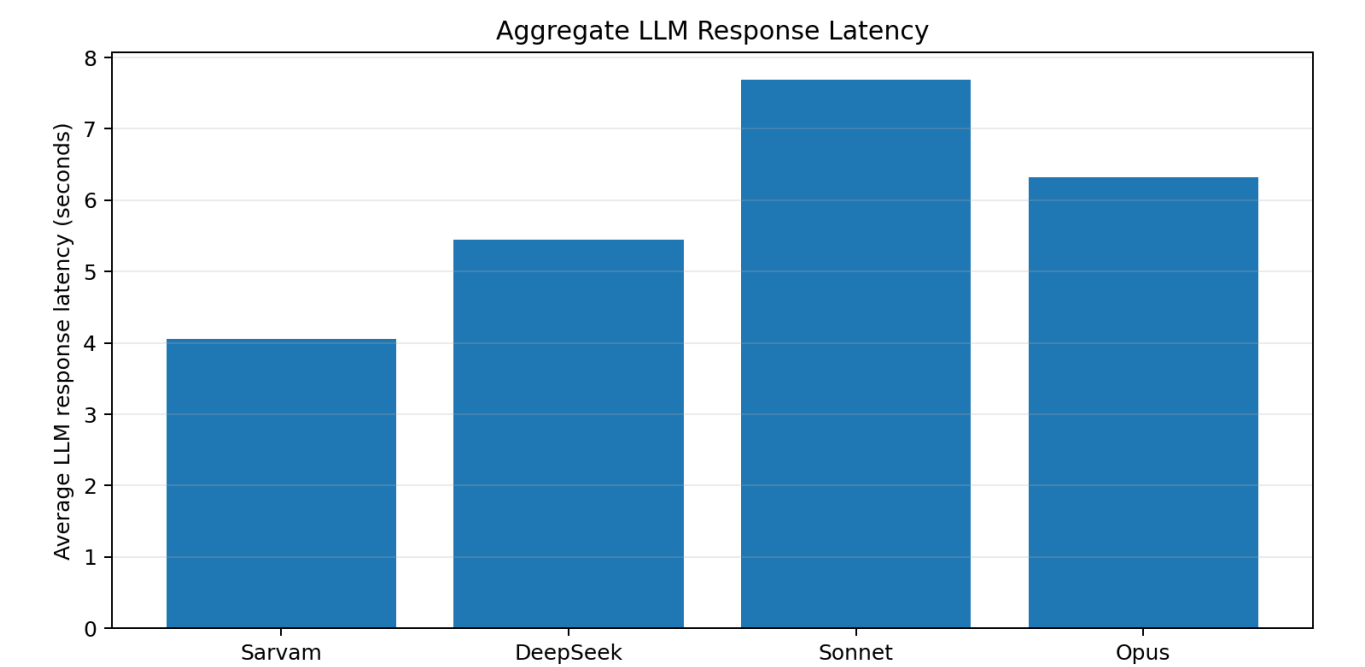
The observation-level error table records actual error categories from the evaluation data. The most frequent current-model categories in the workbook include context errors, factual errors, language-understanding errors, premature confirmation and other behaviour-level issues.

Model	Observed error type	Count
Claude Sonnet	Stage Error	31
Claude Opus	Stage Error	30
DeepSeek	Stage Error	23
Sarvam	Stage Error	22

Sarvam	Context Error	21
Sarvam	Wrong Action	21
Claude Opus	Context Error	18
Claude Sonnet	Context Error	18
DeepSeek	Context Error	18
Sarvam	Factual Error	18
Sarvam	Wrong Intent	18
Claude Sonnet	Wrong Intent	16

7. Latency & Token Usage

Model	N	Avg	P50	P95	Mean Input	Mean Output	Mean Total
Sarvam	133	4.06s	3.10s	7.50s	954	69	1023
DeepSeek	93	5.44s	4.48s	9.99s	1166	83	1249
Claude Sonnet	123	7.69s	6.56s	12.84s	1142	97	1239
Claude Opus	123	6.32s	6.04s	10.66s	1744	110	1854



Latency is LLM/model response latency only. It is not total end-to-end voice latency and excludes STT, TTS, Twilio and infrastructure/network overhead.

8. LLM Cost Comparison

Model	Input tokens	Output tokens	LLM cost/call	1,000 calls	10,000 calls
Sarvam Conversational	1,154	63	₹0.46	₹461	₹4,609
DeepSeek V4.1 Flash	1,168	69	₹0.50	₹497	₹4,973
Claude Sonnet 4.6	1,332	84	₹6.03	₹6,035	₹60,349
Claude Opus 4.8	1,891	99	₹13.70	₹13,698	₹136,975

9. Cost Calculation Method

$$\text{Cost} = (\text{Input} \div 1,000,000 \times \text{Input Price}) + (\text{Output} \div 1,000,000 \times \text{Output Price})$$

For USD-priced models, the USD result is converted to INR using ₹95.71/USD. The reported per-call figures below use the benchmark's separate input/output token measurements.

9. Multilingual Evaluation Evidence

The master workbook contains evidence for English, Hindi, Kannada and mixed-language tracks. It also explicitly distinguishes multilingual capability evidence from production validation.

Language	Model-turns	Avg language understanding*	Avg response-language match*	Interpretation
English	330	100.0%	100.0%	Capability evidence; not production-validated
Hindi	72	85.9%	35.9%	Capability evidence; not production-validated
Kannada	36	93.8%	34.4%	Capability evidence; not production-validated
Mixed	42	97.5%	100.0%	Capability evidence; not production-validated

These are simple summaries across model rows, not standardized language scores. Hindi includes exploratory interactive evidence; the evidence base should not be presented as production multilingual validation.

10. Evaluation Framework Validation

The master workbook records 149 pytest node-level tests as framework validation. These tests are separate from model performance and are not counted as model-turn evaluations.

Test suite	Count	Passed	Failed	Scope
tests/test_deepseek.py	13	13	0	DeepSeek adapter configuration and token handling
tests/test_eval_framework.py	12	12	0	Deterministic evaluator unit tests (facts, grounding, schema)
tests/test_eval_v11.py	17	17	0	LLM-EVAL. 1.1 configuration, availability, Excel summary
tests/test_interactive.py	19	19	0	Interactive session commands, dry-run, isolation
tests/test_master_report.py	6	6	0	Master report aggregation / reconciliation
tests/test_multilingual.py	28	28	0	Multilingual tracks, language match, FakeProvider CLI
tests/test_outbound.py	12	12	0	Outbound opening/stage metrics with FakeProvider
tests/test_outbound_gt_coverage.py	10	10	0	Outbound ground-truth coverage and visit binding
tests/test_session_evaluation.py	9	9	0	Session /evaluate coverage and missing GT handling
tests/test_visit_calibration.py	23	23	0	Visit DATE-before-TIME policy (FakeProvider)
interactive_runs (FakeProvider / DRY_RUN dumps)	56	56	0	Pytest and dry-run session folders
Total collected pytest tests: 149. These MUST NOT be reported as model evaluation trials.				

11. Current Limitations

- The evidence is predominantly offline LLM evaluation; Twilio/real voice-call performance is not validated by this workbook.
- LLM latency excludes STT, TTS, telephony and network overhead.
- LLM cost estimates exclude Twilio, STT, TTS and infrastructure and use explicit pricing assumptions.
- Controlled benchmark and exploratory runs are both present; conclusions should preserve their evidence class.
- Multilingual evidence is limited and is not production validation.
- Historical/out-of-scope model identifiers are retained for traceability but should not be mixed into the current four-model decision set.

12. Recommended Next Steps

- Expand standardized English evaluation to increase the number of controlled observations for the four current models.
- Continue standardized Hindi and Kannada benchmark testing using matched scenarios and consistent ground truth.
- Measure end-to-end voice latency once the Twilio + Lightsail path is enabled, including STT, LLM and TTS.
- Recalculate operating cost using confirmed provider pricing and an agreed USD-INR accounting rate.
- Keep the master workbook as a repeatable evidence layer for future evaluation runs.
- Use observation-level error analysis to prioritize intent, stage, language and grounding improvements.

Appendix — Metric Definitions

Metric	Definition
Intent Accuracy	Share of evaluated turns where predicted intent matches expected intent.
Action Accuracy	Share of evaluated turns where predicted action matches expected action.
Factual Accuracy	Share of evaluated turns judged factually correct against available ground truth/evidence.
Grounded Response %	Share of responses supported by configured project knowledge/context.
Unsupported Claim Rate	Share of responses containing unsupported claims.
P50 / P95 Latency	50th / 95th percentile LLM response latency.
Model-Turn Evaluation	One actual model response evaluated for one customer turn.
Evaluation Run	A unique RUN_ID; exported copies of the same RUN_ID are not additional trials.

Source & Audit Note

Primary source: AI_Calling_Agent_Master_Evaluation_Report.xlsx. The workbook contains the evaluation run inventory, observation-level model results, aggregate metrics, error analysis, latency/token analysis, cost assumptions, multilingual evidence, framework validation and data lineage.